

5 x 7 cm perf board, isolated pads, viewed from the component side. Everything here is SELV: 5 V and SPI only.



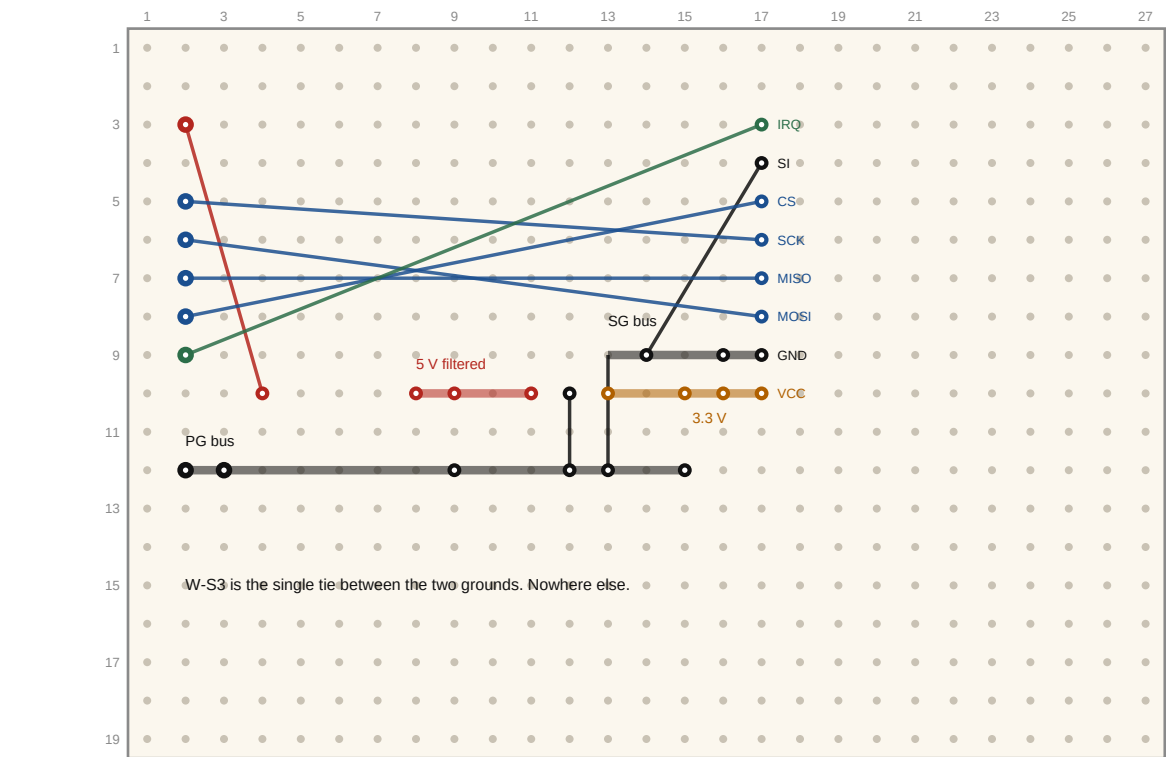
Ref	Part	Holes
R1	100 ohm 1/4 W metal film	(4,10) - (8,10)
C2	47 uF 50 V, EEU-FR1H470	+ (9,10) - (9,12)
U1	MCP1700-3302E, TO-92	VIN (11,10) GND (12,10) VOUT (13,10)
C3	1 uF X7R, C330C105K5R5TA	(15,10) - (15,12)
C4	100 nF X7R, C320C104K5R5TA	(16,10) - (16,9)
M1	SEN-39003 on an 8-pin header	(17,3) .. (17,10), soldered direct

Confirm which end of the SEN-39003 carries the loop antenna before you solder it, and point that end away from the power section.

Its pins are not on a 0.1 in grid and will not fit. Panel jack plus a pigtail, tied down at (1,18).

2. SENSOR BOARD -- point-to-point wiring

X-ray view from the component side. All of this is on the BACK of the board, so it mirrors left-right when you flip it over.



drawn 2:1 -- one grid step = 0.1 in (2.54 mm). Real board 68.6 x 48.3 mm.

5 V 3.3 V GND SPI IRQ

Ref	Net	From	To	Wire
BUS-A	5 V filtered	(8,10)	(11,10)	bare 22 AWG across pads 8-11
BUS-B	3.3 V	(13,10)	(17,10)	bare 22 AWG across pads 13-17
BUS-C	PG ground	(2,12)	(15,12)	bare 22 AWG across pads 2-15
BUS-D	SG ground	(13,9)	(17,9)	bare 22 AWG across pads 13-17
W-S1	5 V in	(2,3)	(4,10)	red 24 AWG
W-S2	LDO GND	(12,10)	(12,12)	black 24 AWG
W-S3	SG-PG tie	(13,9)	(13,12)	black 22 AWG -- the ONLY tie
W-S4	SI strap	(17,4)	(14,9)	black 26 AWG
W-S5	SCLK	(2,5)	(17,6)	26 AWG
W-S6	MOSI	(2,6)	(17,8)	26 AWG
W-S7	MISO	(2,7)	(17,7)	26 AWG
W-S8	CS	(2,8)	(17,5)	26 AWG
W-S9	IRQ	(2,9)	(17,3)	26 AWG

Two grounds, one tie

PG carries the cable's ground return, the bulk cap and the LDO reference. SG carries only the sensor's GND pin, its 100 nF and the SI strap. They meet at exactly one place: W-S3.

Bridge them anywhere else and you have wrapped a ground loop around the LDO; the 100 nF stops being local and the whole point of the split is gone.

How a bus is made on isolated pads

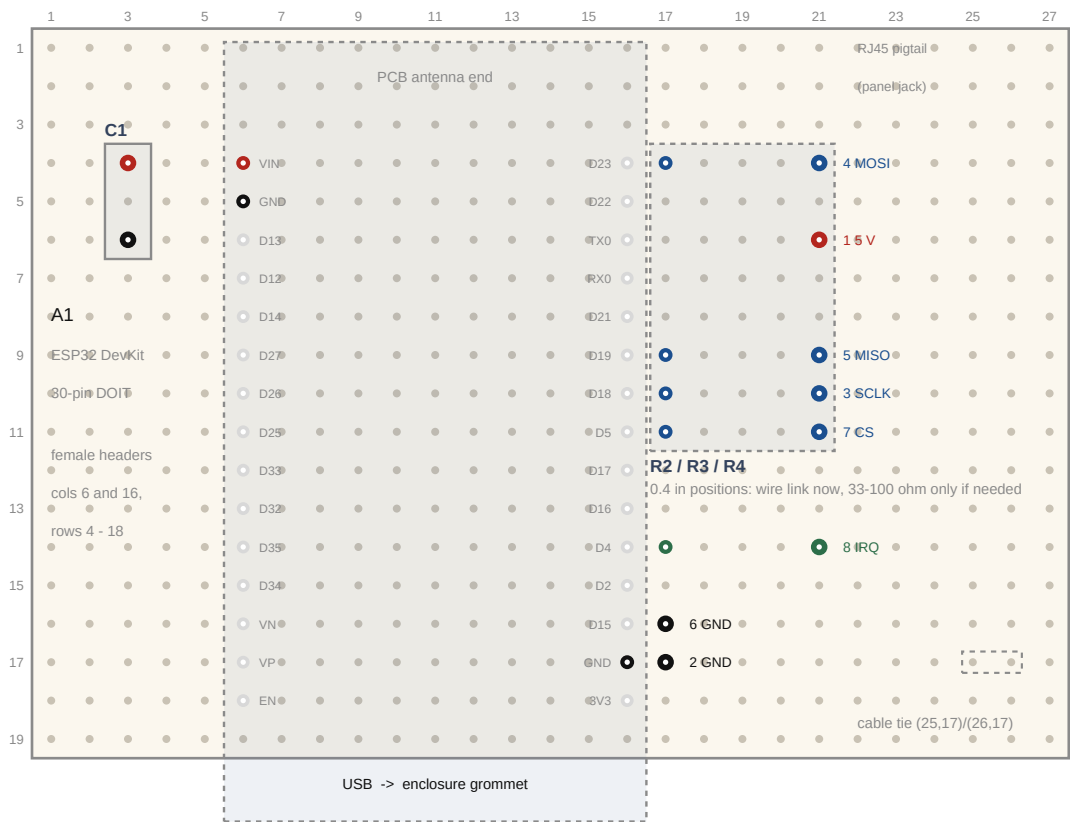
A length of bare 22 AWG laid across the back of the row and soldered to each pad. It does not thread through the holes, which leaves every hole free for a component lead too.

Wire

Kynar 26 AWG wire-wrap for signals: stiff, stays put, strips clean. 22 AWG solid for the buses and power.

3. MAIN BOARD -- component placement

5 x 7 cm perf board. Contents: the ESP32 dev board, the 5 V bulk cap at its pins, and the cable pigtail. Nothing else.



drawn 2:1 -- one grid step = 0.1 in (2.54 mm). Real board 68.6 x 48.3 mm.

5 V 3.3 V GND SPI IRQ

Ref	Part	Holes
A1	ESP32 DevKit, 30-pin DOIT	female headers: col 6 r4-18, col 16 r4-18
C1	470-1000 uF 16-25 V 105 C	+ (3,4) - (3,6) stripe at (3,6)
R2	SCLK series -- wire link	(17,10) - (21,10)
R3	MOSI series -- wire link	(17,4) - (21,4)
R4	CS series -- wire link	(17,11) - (21,11)

Socket this one, unlike the sensor board

Dev boards die, and this one sits metres from the antenna, so the contact-resistance worry that governs the sensor board does not apply. Female headers, and keep BOOT and EN reachable.

Verify your dev board before soldering the headers

The pin names above are a 30-pin DOIT V1 with USB at the bottom. A 36-pin board, or a 0.9 in row pitch, changes everything. Count the pins and measure the row spacing first, then use the dev board itself as the jig so the rows end up parallel.

R2/R3/R4 are wire links on day one

Only the three lines the ESP32 drives get a position. MISO and IRQ are driven from the far end, so series damping here would do nothing. Fit 33-100 ohm only if the 3 m sweep in README 16 misbehaves.

The overhang goes at the USB end

The dev board is about 3 mm longer than the perf. Hang that off the USB end so the plug clears the board edge.

X-ray view from the component side. Back-side wires pass under the perf board, not under the ESP32, so a long run costs nothing.



Ref	Net	From	To	Wire
W-M1	C1+ to VIN	(3,4)	(6,4)	red 22 AWG, keep under 10 mm
W-M2	C1- to GND	(3,6)	(6,5)	black 22 AWG, keep under 10 mm
W-M3	MOSI stub	(16,4)	(17,4)	26 AWG
W-M4	MISO stub	(16,9)	(17,9)	26 AWG
W-M5	SCLK stub	(16,10)	(17,10)	26 AWG
W-M6	CS stub	(16,11)	(17,11)	26 AWG
W-M7	IRQ stub	(16,14)	(17,14)	26 AWG
W-M8	IRQ to cable	(17,14)	(21,14)	wire link, no resistor
W-M9	MISO to cable	(17,9)	(21,9)	wire link, no resistor
W-M10	5 V to cable	(6,4)	(21,6)	red 22 AWG, long run is fine at 1 mA
W-M11	GND to ESP32	(17,17)	(16,17)	black 22 AWG
W-M12	GND link	(17,16)	(17,17)	black 22 AWG

470-1000 uF, 105 C, on the VIN and GND pins with under 10 mm of wire either side. It is the reservoir the USB cable cannot supply fast enough during a WiFi burst -- README 7.4. Watch the polarity: the stripe is the minus lead, at (3,6).

Same T568B assignment as README 7.1. Only the two grounds move: they land at (17,16) and (17,17) so they reach the ESP32's right-hand GND pin in one hop.

Route the USB cable away from the Cat5. Do not bundle them.

5. Build order, and the things that will bite

Read this before you cut anything. Sections refer to the project README.

Sensor board

1. Cut/snap the perf to 27 x 19 holes and de-burr. Mark hole (1,1) in the corner with a pen; every coordinate here counts from it.
2. Solder the four bus wires first, while the board is still flat and empty: PG (row 12, cols 2-15), SG (row 9, cols 13-17), 5 V filt (row 10, cols 8-11), 3.3 V (row 10, cols 13-17).
3. R1, then C2, then U1, then C3, then C4. Shortest parts first.
4. W-S3, the single SG-PG tie. Do it deliberately and mark it.
5. Solder the 8-pin header to the SEN-39003, then solder that assembly into (17,3)..(17,10). Check the antenna end points right, away from the power section, before you commit.
6. W-S4, the SI strap. Without it the part talks I2C and nothing works -- README 5.
7. The five signal wires and the pigtail. Cable tie at (1,18).
8. Before power: ohmmeter from 3.3 V to either ground. Anything under a few hundred ohms means a bridge; find it now.

Main board

1. Count your dev board's pins and measure its header row spacing. Only then solder the female headers, using the dev board itself as the jig so the rows end up parallel.
2. C1 with the stripe at (3,6), then W-M1 and W-M2. Short.
3. Stubs, links and the pigtail. Cable tie at (25,17).
4. Power it with no ESP32 fitted and confirm 5 V appears at the cable's pin 1 and nowhere it should not.

Things that will bite

The X-ray thing. Both wiring pages are drawn as if you could see through the board from the top, because that is how you place parts. When you flip the board to solder, left and right swap. Getting this wrong is the single most common perf-board error.

Isolated pads, not strips. If what you actually have is stripboard, every one of these rows is already a bus and the layout is wrong -- you would need track cuts. Check before you buy.

RJ45 on 0.1 in perf. It does not fit; the pins are staggered off-grid. Panel jack and pigtail, both boards.

This jack is not Ethernet. It carries 5 V and SPI. A live PoE port puts 48 V on those lines and destroys both ends. Label both boxes now, while you are holding them -- README 7.1.

Nylon standoffs, nylon screws, on the sensor board especially. A steel screw near a 500 kHz loop antenna is a shorted turn.

Rigidity is a measurement, not a feeling. README 11.3: the breadboard's noise floor fell by two thirds the moment the build was handled. The Phase 2 test in README 15 is survey, handle the box, survey again. If the rates move, the build is furniture and nothing measured on it means anything.

Then set data_rate: 200kHz in lightning-detector.yaml before the first survey -- it defaults to 1 MHz, README 7.1.



2.000 in / 50.8 mm -- 20 holes at 0.1 in pitch. If this does not measure true, your printer scaled the page.